BankSystem

Infrastructure and Architecture

An overview for software architects · November 2026

A web bank that connects to other banks through a central hub, a sample shop that takes card and direct payments through it, and one central sign-in for both. Each part is an independent Docker Compose stack.

3 stacks
kyc, bank, shop, deployed on their own

3 ports
8080 Keycloak, 8081 bank, 8082 shop

RSA + AES
on every bank to hub call

1 sign-in
OpenID Connect through Keycloak

60 / min
public API calls per client IP

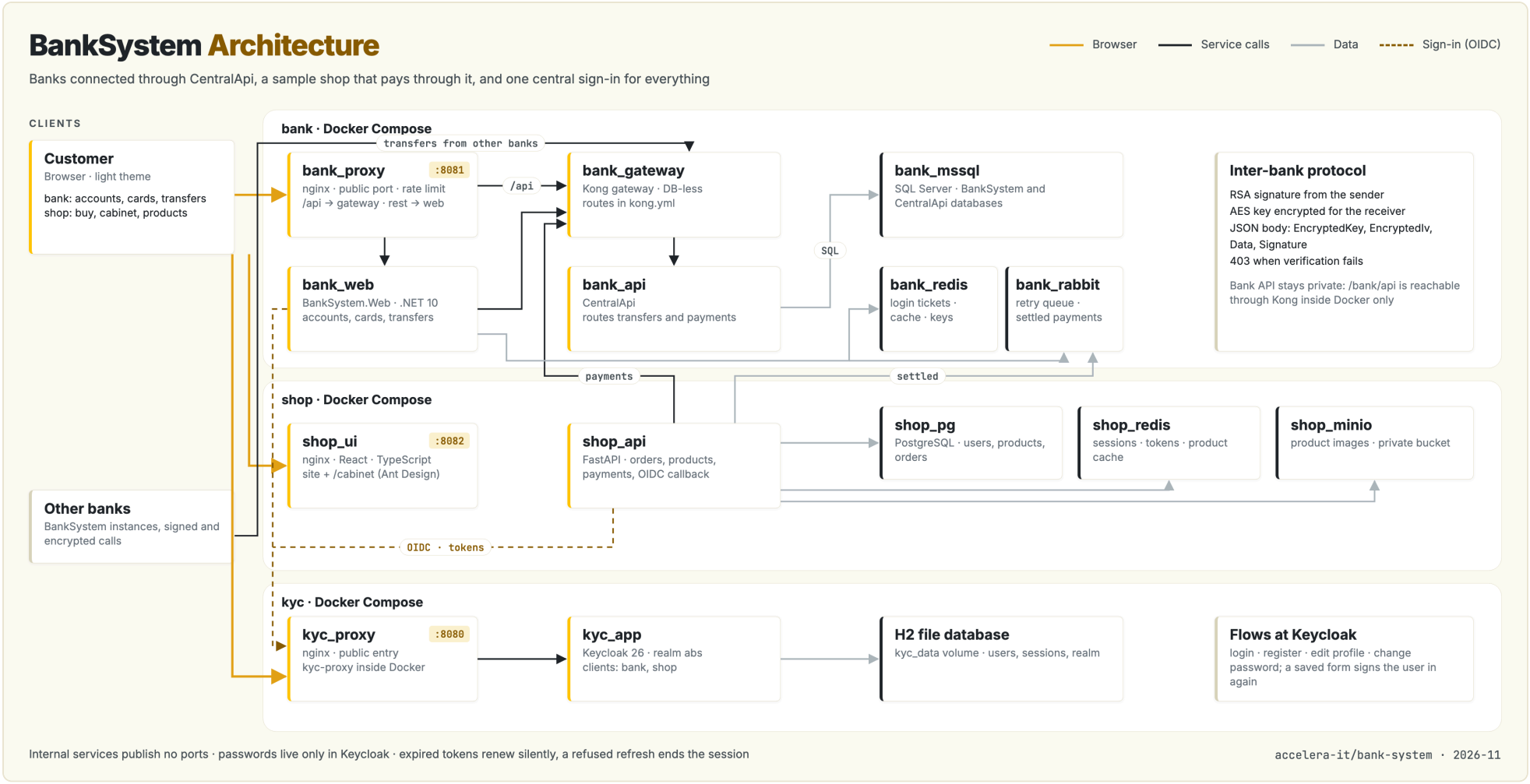
.NET 10
with Python, React, TypeScript

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A Architecture

The whole system on one page.



1 Goals and Key Properties

Banks stay independent; the hub and the sign-in are shared.

Property	How it is ensured
One account everywhere	Keycloak holds every password. The bank and the shop have no login or register pages and create their local user on first sign-in, by email.
Banks stay independent	Each bank has its own database, RSA key and identity. CentralApi only routes between them, by name, swift code and country.
Bank traffic is unreadable and signed	Every call is hybrid encrypted with AES and an RSA-wrapped key, and signed by the sender. Verification failures answer 403.
Sessions follow Keycloak	Each app keeps the refresh token. Expired tokens are renewed; a refused refresh ends the session at once.
A credential change signs out	A saved profile or password ends the app session and the Keycloak session, so the user signs in again.
Small public surface	Only nginx in each stack publishes a port. Databases, Redis, MinIO, the gateway and the apps are internal.
Public API calls are limited	nginx allows each client IP 60 calls a minute on `/api/`, with short bursts. Internal calls do not pass through it and are not limited.
Money moves once	A transfer is saved in one database transaction, and a delivery with no clear answer is repeated until there is one. A bank credits each reference number once.
Keycloak being away signs nobody out	A network failure during refresh leaves the session alone, and the next request tries again.

2 System Context

Who uses the system and what it depends on.

Actor	Reaches	Through
Customer	Bank pages, shop site and cabinet	bank_proxy :8081, shop_ui :8082
Customer signing in	Login, register, profile and password forms	kyc_proxy :8080
Bank staff	All accounts, transfer lookup, pending transfers	Administration area of the bank
Merchant (shop)	Direct and card payments	bank_gateway (Kong)
Other banks	Global transfers and card payments	CentralApi, signed and encrypted
Keycloak root user	Realm administration	kyc_proxy :8080/admin; kyc_app makes it permanent

3 Components

Three stacks, each with its own Compose file and env file.

Stack	Service	Technology	Role
bank	`bank_proxy`	nginx	Public entry: `/api/` to the gateway, the rest to the bank; rate limit
bank	`bank_web`	ASP.NET Core 10	Accounts, cards, transfers, payments, administration
bank	`bank_gateway`	Kong, DB-less	Gateway for every API call, routes in `kong.yml`
bank	`bank_api`	ASP.NET Core 10	CentralApi: routes transfers and payments between banks
bank	`bank_mssql`	SQL Server 2022	BankSystem and CentralApi databases
bank	`bank_redis`	Redis	Login tickets, cache, Data Protection keys
bank	`bank_rabbit`	RabbitMQ	Delivery and retry queues for pending transfers, settled payments for the shop
shop	`shop_ui`	nginx, React, TypeScript	Shop site and cabinet, proxies `/api/`
shop	`shop_api`	FastAPI, SQLAlchemy	Products, orders, payments, OIDC callback
shop	`shop_pg`	PostgreSQL	Users, products, orders
shop	`shop_redis`	Redis	Sessions, tokens, product cache
shop	`shop_minio`	MinIO	Product images in a private bucket
kyc	`kyc_proxy`	nginx	Public entry to Keycloak
kyc	`kyc_app`	Keycloak 26, H2	Sign-in, realm `abs`, clients `bank` and `shop`

4 Startup Order and Ports

Each stack joins the networks of those before it.

Order	Stack	Public port	Creates	Joins
1	kyc	8080	`kyc_lan`	—
2	bank	8081	`bank_lan`	`kyc_lan`
3	shop	8082	`shop_lan`	`bank_lan`, `kyc_lan`

Inside the containers the apps listen on 5000, the bank's nginx on 80 and Keycloak on 8080. MinIO and Keycloak refuse _ in a host name, so they are reached through the `shop-minio` and `kyc-proxy` network aliases.

5 Sign-In

OpenID Connect authorization code flow, with PKCE in both apps.

1. The bank (`/account/login`) or the shop (`/api/auth/login`) sends the browser to Keycloak. Register adds `prompt=create`.
2. Login asks for email and password. Register asks for Full Name, email, password and confirmation, and its button waits for the agreement checkbox.
3. Keycloak returns a code to `/signin-oidc` or `/api/auth/callback`, and the app exchanges it through `kyc-proxy`.
4. The app checks the issuer, audience, nonce and expiry, creates its local user on the first sign-in and takes the Full Name from Keycloak.
5. The session is stored in Redis, with the tokens, and the browser only holds a cookie that points to it.

Browsers use `localhost:8080` and the apps use `kyc-proxy:5000`. Keycloak's dynamic backchannel keeps the issuer the same for both.

6 Sessions and Tokens

What happens after sign-in.

Event	Behaviour
Access token expires	The next request renews the tokens with the refresh token, with no sign to the user.
Keycloak refuses the refresh	The session is deleted and the user is sent to the login page. This covers logout elsewhere and blocked users.
Keycloak is unreachable	The session stays and the next request tries again.
Logout	The session ends and the browser goes to Keycloak's logout, which ends the shared sign-in too.
Second app in the same browser	Keycloak's session signs the user in with no form.

The bank does this in `KycTokenRefresher`, on cookie validation. The shop does it in `current_user`, on every request.

7 Edit Profile and Change Password

Forms run at Keycloak, in a popup.

1. The menu item opens a popup through the login flow with `kc_action`: `UPDATE_PROFILE` or `ABS_UPDATE_PASSWORD`. Neither app has a settings page.
2. Edit profile shows Full Name, Phone and Address, with the email read-only.
3. Change password asks for the current password, the new one and its confirmation. The `abs` theme adds the field and a Keycloak provider checks it on the server.
4. A saved form returns `kc_action_status=success`: the app session and the Keycloak session end, and the app behind goes to login.
5. A cancelled form changes nothing: the popup closes and the app behind reloads.

8 Inter-Bank Security Protocol

Every call between a bank and CentralApi.

Step	Detail
Sender signs	With its own RSA private key
Sender encrypts	The payload with a new AES key and IV, and those with the receiver's RSA public key
Message	Base64 JSON: `EncryptedKey`, `EncryptedIv`, `Data`, `Signature`
Receiver verifies	The `[DecryptAndVerifyRequest]` filter decrypts, verifies and replaces the action's first argument with the JSON
Failure	403, and the action never runs

Keys use the .NET `<RSAKeyValue>` XML format; `docs/rsa-keygen.sh` generates them.

9 Routing

What nginx and Kong send where.

Public path	Internal path	Target
`:8081/*`	same	`bank_web`
`:8081/api/pay`	`/pay`	`bank_api`, with `X-Forwarded-Prefix: /api` from nginx
`:8081/api/pay/continue`	`/pay/continue`	`bank_api`, with the same prefix
`:8081/api/static/*`	`/static/*`	`bank_api`: fonts and Bootstrap of the pay pages
`:8081/api/swagger/*`	`/swagger/*`	`bank_api`, Development only
`:8081/api/*`	`/api/*`	`bank_api`
Internal only: `bank_gateway:5000/bank/api/*`	`/api/*`	`bank_web`; nginx does not publish it

`/pay` exists in the bank and in CentralApi, so CentralApi's page is public as `/api/pay`.

10 Payment Flows

The two ways a shop gets paid.

Step	Direct payment	Card payment
1	The shop signs a request and the browser posts it to `/api/pay`	The cabinet sends the card to `shop_api`
2	CentralApi lists the banks and the customer picks one	`shop_api` calls CentralApi through the gateway
3	The bank's `/pay` page confirms, signed in through Keycloak	CentralApi finds the bank by the card's first 3 digits
4	The transfer goes bank → gateway → CentralApi → gateway → bank	The bank checks the card and moves the money
5	The bank posts a signed result to the shop, which marks the order paid	The shop marks the order paid, or Processing if the answer is not final

The receiving bank checks that the recipient's name equals the account owner's Full Name, so `RECIPIENT_NAME` in `shop.env` must match the shop's bank user.

11 Reliable Delivery

How money moves once, even when a call gets no answer.

Case	What happens
Internal transfer	Both sides are one `SaveChanges`. `Balance` is a concurrency token, so parallel requests cannot overdraw an account
Global transfer	The money leaves the account together with a `PendingTransfers` row, in one transaction, so a crash cannot lose it
Answer 2xx	Delivered: the row ends
Answer 4xx	Refused: the money returns and the row ends, in one transaction
5xx, 408, 429, timeout	Unknown: the money stays held. Up to 3 tries in the request, then the `transfers.deliver` queue, which `transfers.retry` refills every 30 seconds, up to 1440 attempts
Still no answer	A critical log, and Retry and Refund buttons at `/administration/pendingtransfers`. Refund only after the other bank confirms the money did not arrive
Lost message	`PendingTransfers` is the record and RabbitMQ only speeds it up: `PendingTransferRetryService` picks up anything older than 5 minutes, also with no broker
Repeated delivery	The receiver credits a reference number once per account, with a filtered unique index behind the check
Merchant with no final answer	`shop_api` gives each attempt a random reference number and asks `POST /api/CardPayments/status`: when the bank's message arrives on `shop.payments`, on orders and payments, and every 5 minutes. The order is Processing and cannot be paid again

The state is Pending, Completed, Refunded or NotFound. CentralApi asks every bank, because the shop does not know the customer's bank for a direct payment. The shop's message is only a hint and it can only read its own queue, so a false message changes nothing.

12 Product Images

Uploaded in the cabinet, served by the API.

Rule	Detail
Storage	A private MinIO bucket, `products`; only `shop_api` reaches it
Upload	Signed-in users, 5 MB at most, through `POST /api/products/images`
Type	PNG, JPEG or WebP, taken from the bytes and not from the file name
Serving	`GET /api/products/images/{key}`, cached for a year, with `nosniff`
Editing and deleting	In the cabinet's product list. A product with orders cannot be deleted, and a deleted product's image stays in the bucket

13 Look and Assets

One light look, served locally.

Area	Detail
Palette	The light colors of cbu.uz in the bank, the shop and the Keycloak theme: gold #ffc700 and #e4a216, deep gold #8a5a00 and cream #f6f5e7. There is no dark mode.
Fonts	Inter, in `wwwroot/fonts`, `shop/app/public/fonts` and CentralApi's `wwwroot/static`
Icons	Font Awesome in the bank, Ant Design icons in the cabinet and inline SVG in the shop site
Styles	Bootstrap in the bank and CentralApi, Tailwind in `shop/app/public/js/tailwind.js`
CDNs	None: browsers load nothing from outside the stacks. Building the images still needs the registries.
Bank menu	Overview, Transfers, Cards, Administration and Settings, with the Full Name and a red Log out below
Shop	A landing page and a `/products` catalog with no detail page; the cabinet has Overview, Shopping and Settings

14 Data Storage

Where each piece of data lives.

Store	Holds	Volume
SQL Server	Users, accounts, cards, transfers, pending transfers, known banks	`bank_mssql_data`
Redis (bank)	Login tickets with tokens, cache, Data Protection keys	`bank_redis_data`
RabbitMQ	Delivery, retry and settled payment messages	`bank_rabbit_data`
PostgreSQL	Shop users, products, orders, open payment attempts	`shop_pg_data`
Redis (shop)	Sessions, tokens, login state, product cache	`shop_redis_data`
MinIO	Product images	`shop_minio_data`
H2	Keycloak users, credentials, sessions, realm	`kyc_kyc_data`

The bank and CentralApi migrate with EF Core on startup. The shop creates its tables in `seed.py`. The `abs` realm is imported on first start only.

15 Security

Controls, and what is deliberately left for a real deployment.

Area	Control
Passwords	Only in Keycloak; the bank and the shop store none
Change password	The `ABS_UPDATE_PASSWORD` required action checks the current password on the server, not only in the form
OIDC	State, nonce and PKCE; the client secret stays on the server
Redirects	`next` and `returnUrl` stay on the site; Keycloak allows only registered URIs
Sessions	HttpOnly cookies that point to Redis, with no tokens in the browser
Rate limit	60 calls a minute per IP on public `/api/` in nginx, answered with 429
Uploads	Type from the bytes, private bucket, `nosniff`
Messages	RabbitMQ publishes no port, the `shop` user can only read its own queue, and the shop treats a message as a hint

16 Testing

What runs, and on what.

Suite	Technology	Runs on
Bank and CentralApi (4 projects)	xUnit, FluentAssertions, Moq	EF Core InMemory, `WebApplicationFactory`
Shop API	pytest	SQLite, fake Redis, fake storage
Shop UI	Playwright, TypeScript	The whole stack, real Keycloak, CentralApi stub
Keycloak	JUnit, Java	Real Keycloak on H2: login, register, tokens, sessions

SQL Server and RabbitMQ are not covered by the test suites: they are checked by running the real stack.

17 Operations

Running it.

```
docker compose -f kyc/docker-compose.yml up -d --build
docker compose -f bank/docker-compose.yml up -d --build
docker compose -f shop/docker-compose.yml up -d --build

cd bank && dotnet test
docker compose -f shop/docker-compose.test.yml run --rm --build shop_api_tests
docker compose -f shop/docker-compose.test.yml up --build --abort-on-container-failure --exit-code-from shop_ui_tests shop_ui_tests
docker compose -f kyc/docker-compose.test.yml up --build --abort-on-container-failure --exit-code-from kyc_tests kyc_tests
```

Account	Password	Use
demo@abs.uz	123456	Default user: bank administrator with 10,000,000 UZS
admin	123456	Keycloak root user

18 Risks and Roadmap

What a real deployment still needs.

Risk	Now	Next
Secrets in a public repository	Env files with development values are committed	Rotate every value and move them to a secret store
Email is not verified	Users are matched by email, so anyone can claim an unused address	Turn on Keycloak's `verifyEmail`, with SMTP
H2 for Keycloak	Single instance, development use only	Move to a production database
Plain HTTP	Everything runs on localhost	Terminate TLS at nginx and require it in Keycloak
SQL Server on Apple Silicon	Runs under amd64 emulation	Run it natively on the servers
Recipient name check	Full Name comes from Keycloak on each sign-in, so renaming the shop's bank user stops its payments	Match accounts by id, not by name
Mixed bank versions	A bank without the once-only check could credit a repeated delivery twice	Update every connected bank before enabling retries
Orphaned images	A deleted product leaves its image	Delete the object with the product